

Know Sure Thing (KST)

Indicators & Analysis / Momentum & Oscillators | advanced | 5 min read | DeepCharts Help Center | July 16, 2026

The Know Sure Thing (KST) is a momentum oscillator based on multiple smoothed Rate of Change (ROC) calculations. It combines short-, medium- and long-term momentum into one indicator and includes a signal line for crossover analysis.

Where a single Rate of Change (ROC) reading only tells you about one lookback window, KST blends four of them — each individually smoothed and weighted — so a genuine momentum shift has to show up across several horizons before the line turns. That makes it slower than a raw ROC but considerably harder to whipsaw.

What it is

KST answers the question: is momentum turning across multiple timeframes at once, or only on one? It plots two lines in a separate panel:

- **KST** — the composite momentum line built from four smoothed ROC components.
- **Signal** — a smoothed copy of KST used for crossover signals.

The zero line separates bullish from bearish momentum bias. The indicator is widely credited to Martin Pring.

[SCREENSHOT PLACEHOLDER] Price chart with the KST indicator in a lower panel, KST and Signal lines in contrasting colors, a bullish KST-over-Signal crossover below the zero line highlighted `kst-on-chart.png`

When to use it

- Identifying momentum shifts that are confirmed across short, medium and long lookbacks simultaneously.
- Confirming a trend — KST holding above zero supports a bullish bias, below zero a bearish one.
- Spotting divergences between KST and price at swing highs and lows.
- Flagging overextended conditions when KST stretches far from the zero line.

Quick start

1. Open a price chart — see [Open Your First Chart](#).
2. Click the green **Indicators** button (or press **Ctrl I**) to open the **Indicator List**.
3. Search for **Know Sure Thing** and click **+** to add it. It appears in its own panel below the price.
4. Click the gear icon to open its settings.
5. A widely used starting configuration is Martin Pring's classic setup: ROC lengths of 10, 15, 20 and 30, average lengths of 10, 10, 10 and 15, and a signal period of 9. Adjust from there for your timeframe.

[SCREENSHOT PLACEHOLDER] KST settings dialog open showing the General group with Use percent, the Average group with Average Type, Avg Length 1-4 and Signal Period, and the ROC group with ROC Length 1-4 `kst-settings-dialog.png`

How to read it

The core reads, from the indicator's own logic:

Observation	Interpretation
KST above zero	Bullish momentum bias
KST below zero	Bearish momentum bias
KST crossing above Signal	Potential bullish shift

Observation	Interpretation
KST crossing below Signal	Potential bearish shift

The strongest signals combine both: a KST/Signal crossover that also crosses the zero line, or a crossover in the direction of the side of zero you are already on (a bullish cross above zero is a continuation signal, not a reversal call). Divergence — price making a new extreme while KST does not — warns that the multi-timeframe momentum behind the move is fading.

Settings reference

Grouped as in the indicator's configuration window.

Chart area

Setting	What it does
Horizontal	Displays KST in a separate panel.
Vertical	Displays the indicator vertically (if supported).
Panel selector	Chooses which panel the indicator appears in.
Use sec. axis	Enables secondary scaling.

Parameters — General

Setting	What it does
Use percent	Switches the ROC calculation between raw values and percentage-based momentum. Percentage mode makes readings comparable across instruments with different price scales.

Parameters — Average

Setting	What it does
Average Type	Defines the smoothing method applied to each ROC component.
Avg Length 1–4	Controls the smoothing of each individual ROC calculation. Higher values produce a smoother but slower response.
Signal Period	Defines the smoothing length of the signal line used for crossover signals.

Parameters — ROC

Setting	What it does
ROC Length 1–4	Defines the lookback period for each Rate of Change component. Shorter lengths react faster; longer lengths capture broader momentum trends.

Levels

Setting	What it does
Middle Level	Defines the neutral (zero) reference line.
Middle Level Color	Customizes the zero-line color.
Level Width	Adjusts the thickness of the reference line.

Subgraphs

- **KST** — controls the appearance of the main momentum line: color, line style, width, auto-color behavior and label display.
- **Signal** — controls the appearance of the signal line: color, line style, width and label display.

Display options available for both lines: **Name Label**, **Value Label**, **Name Background**, **Value Background**, **Include on Auto Center**, and **Chart color for marker**.

Tips and common mistakes

- **Keep the four ROC lengths staggered.** The indicator's value comes from blending different horizons; setting the lengths close together collapses it into an expensive single ROC.
- **Signal crossovers lag by design.** KST is built from smoothed components, so crossovers confirm turns rather than predict them — pair it with structure or level analysis for entry timing.
- **Zero-line context first, crossover second.** A bearish crossover far above zero often marks only a pause in an uptrend, not a reversal.
- **Scale the lengths to your chart.** Pring's classic values were designed around longer-term analysis; intraday traders typically shorten all four ROC and average lengths proportionally.

Related articles

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